SharePoint Online End-to-End Lab 2026

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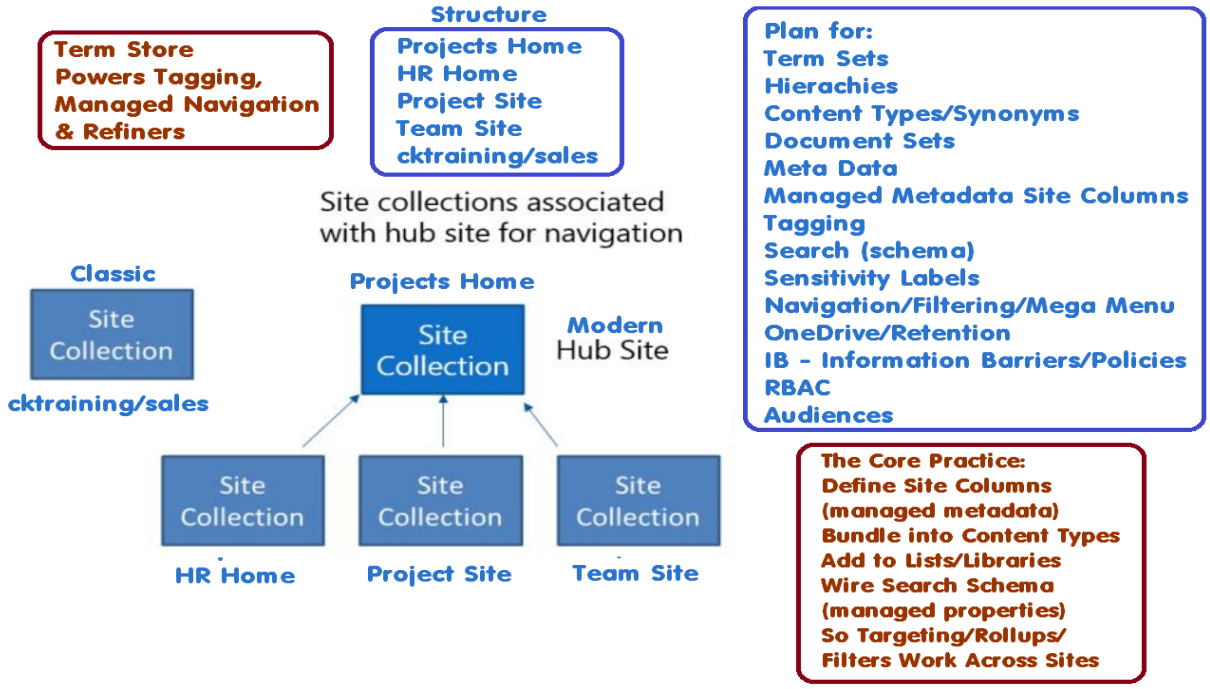

- Starts with **modern sites + a hub**,
- Includes one **classic site + subsite** to modernize,
- Walks students through the **core practice** (taxonomy → content types → libraries → search schema),
- Layers on **audience targeting & navigation**,
- Finishes with **Power Apps + Power Automate**, **OneDrive/IRM/sync** settings, and optional **Information Barriers**.

AdventureWorks flavored, aligned the order to how SPO actually works in 2026, and added "Why This Matters" notes with links to official guidance.

SharePoint Online End-to-End Lab

Title: From Information Architecture to Apps: Taxonomy, Search & Hub Navigation
Scenario: *AdventureWorks Bicycles*
Time: ~4–6 hours total (modular)
Audience: Admins / Power users / Makers
Prereqs: SharePoint Admin, Global Admin or equivalent delegated permissions; Power Apps / Power Automate maker permissions

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Module 0 — What students will create

Sites & topology

- **Projects Home** (modern team site) → becomes **Hub**
- **HR Home** (modern site) → associated to hub
- **Project Site** (modern site) → associated to hub
- **Team Site** (modern site) → associated to hub
- **cktraining/sales** classic site **with a subsite** → used to practice modernization patterns

Platform capabilities

- Managed **Term Store** (taxonomy), content types, and **managed properties** for search refiners/rollups. [sharepointdiary.com], [c-sharpcorner.com]
- **Hub navigation** + **audience-targeted links** and a **Highlighted Content** rollup using KQL. [pragmaticworks.com], [cdm.iamcloud.info], [learn.microsoft.com]
- **Power Apps** form + **Power Automate** flow integrated with a SharePoint list.
- **OneDrive/SharePoint admin** settings: retention, storage, sync restrictions, **IRM** (optional), and notes on **Information Barriers**. [dominiquehermans.com], [rishandigital.com]

Module 1 — Create the sites and register a hub

Why: Modern, flat site collections + hub associations is Microsoft's recommended IA (not subsites). Hubs unify branding/search/nav across related sites. [pragmaticworks.com]

1. **Create modern sites** (Admin center → SharePoint → *Active sites* → **Create**):
    - **Team site**: *[YourName] – Projects Home*
    - **Team site**: *[YourName] – HR Home*
    - **Team site**: *[YourName] – Project Site*
    - **Team site**: *[YourName] – Team Site*

2. **Register a hub** on *Projects Home* (PowerShell):

PowerShell

Connect-SPOService -Url https://<tenant>-admin.sharepoint.com

Register-SPOHubSite https://<tenant>.sharepoint.com/sites/<YourName>-ProjectsHome -Principals $null

Show more lines

Notes: Register-SPOHubSite promotes an existing site to a **hub**; later you can grant who may associate sites. Limit: up to 2,000 hubs/tenant. [Refactor_S...Admin_2026 | Word]

3. **Associate sites** to the new hub (Admin center → select site → Hub → **Associate with a hub** → choose *Projects Home*). Hub nav now appears across associated sites. [pragmaticworks.com]

4. **Create one classic site + subsite**:
    - Create site collection **cktraining** (classic template) and a **sales** subsite. We'll use this later to discuss modernization path (and why not to build new subsites). [pragmaticworks.com]

Module 2 — Core practice: build taxonomy → content types → libraries → search schema

Why: Terms + content types standardize tagging. Mapping to managed properties unlocks refiners, sorting, and KQL for rollups. [sharepointdiary.com], [c-sharpcorner.com]

Step A — Create term sets (Tenant-wide Term Store)

1. SharePoint Admin Center → **Content services** → **Term store**.
2. Create a **group** *AdventureWorks*; then **term sets**:
    - **Department** (e.g., HR, Finance, Sales)
    - **RecordType** (Policy, Contract, Procedure)
    - **Region** (NA, EMEA, APAC)
      Tip: Define synonyms for common alternates (e.g., "Policy" ~ "Standard"). [sharepointdiary.com]

Step B — Create managed metadata site columns & content types

1. **Site columns** (type **Managed Metadata**) bound to the Term Store sets:
    - Department → term set Department
    - RecordType → term set RecordType
    - Region → term set Region [mrsharepoint.com]

2. **Content types**:
   - **AW Policy** (Document): add Department, Region
   - **AW Contract** (Document): add Department, Region, RecordType
     Note: Content types are the reusable schema unit across libraries. [sharepointdiary.com]

Step C — Add content types to libraries across the sites

1. On **HR Home** and **Project Site**, create libraries **Policies** and **Contracts**; enable "Allow management of content types" → add **AW Policy** or **AW Contract** respectively.
2. Upload a few files and tag them with the new terms.

Step D — Map metadata to search managed properties

1. SharePoint Admin Center → **More features** → **Search** → **Manage Search Schema**.
2. Find crawled properties for your columns (ows_taxid_<FieldName> for managed metadata). Map to **RefinableString** or create a new **managed property**; mark **Searchable/Queryable/Refinable/Sortable** as needed. [c-sharpcorner.com]
3. **Reindex** the library (Library settings → Advanced settings → Reindex), then wait for index to refresh.

Checkpoint: If a column isn't mapped to a useful **managed property**, you can't filter/refine it globally (e.g., in Highlighted Content or search refiners). [c-sharpcorner.com]

Module 3 — Navigation, audience targeting, and rollups

Why: Modern SharePoint supports **local**, **hub**, and **global** navigation (app bar). You can target nav links and web parts by **audience** for relevance (not security). [pragmaticworks.com], [helpcenter...zensai.com], [cdm.iamcloud.info]

1. **Hub navigation**
   - On the **hub site** (Projects Home) → **Edit** hub navigation.
   - Add links (HR Home, Project Site, Team Site, Sales) and **enable audience targeting** for links meant for managers vs. all staff. [pragmaticworks.com], [cdm.iamcloud.info]
2. **Global navigation** (optional in this lab)
   - SharePoint app bar → enable & configure tenant-wide links to the intranet home and help resources. [helpcenter...zensai.com]
3. **Highlighted Content Web Part (HCWP) rollup**

- On the hub home page, add **Highlighted Content**, set **Source** = *This hub* (or *All sites*), **Custom query (KQL)**:
- (ContentTypeId:0x0101* OR FileType:docx) AND
- (RefinableString00=Policy OR RecordType=Policy) AND
- Department=HR
- Switch to a **Grid/Compact** layout; enable **audience targeting** if needed. HCWP supports layout, filters, and KQL for precise targeting. [learn.microsoft.com]

Tip: Audience targeting personalizes; it doesn't restrict access. Always secure content with permissions. [cdm.iamcloud.info]

Module 4 — Working with large libraries

Why: Metadata Navigation and Filtering adds a left-hand tree + key filters, and SharePoint auto-indexes columns for performance. [en.ittrip.xyz]

1. Library settings → **Metadata navigation settings** → choose **Navigation hierarchy** (e.g., RecordType, Department) and **Key Filters** (e.g., Region).
2. Ensure **automatic indexing** is enabled. [en.ittrip.xyz]

Module 5 — People data that actually works in search (optional)

Why: Binding **user profile properties** to term sets improves consistency and people search; then map the crawled property to a **managed property**. [sympraxisc...ulting.com], [mrsharepoint.com]

1. Classic **Manage User Properties** → New property Skills (or reuse) → **Configure a Term Set** (e.g., Skills term set).
2. Set privacy to **Everyone** so the property is crawlable; map its crawled property to a managed property in the search schema to query/refine. [mrsharepoint.com]

Module 6 — Classic site touchpoint: modernization discussion

Why: Classic sites/subsites still exist but modern experiences (flat + hubs) are the strategic path; some features (e.g., app bar) don't apply to classic. [pragmaticworks.com], [helpcenter...zensai.com]

1. Review the **cktraining** classic site + **sales** subsite created in Module 1.
2. Show differences in UI and where classic limits modern experiences (no app bar, etc.). [helpcenter...zensai.com]

3. Explain options to **replace** with modern sites and associate to a hub rather than creating new subsites (use **Site Swap** for roots; don't delete). [m365admin....sontek.net]

Module 7 — Power Apps + Power Automate (maker track)

Why: SPO lists are first-class data sources for **Power Apps** and **Power Automate**—great for form UX and event-driven processes.

1. In **cktraining/sales**, create list **Sales Commission Payments** (your existing lab data works).
2. Click **Integrate → Power Apps → Create an app**; rename to *Submit your commission*; adjust the gallery to show stakeholder **DisplayName**.
3. Click **Integrate → Power Automate → Create a flow**
   - Start with template "When item is created…" → replace email with **Teams: Post message as Flow bot to recipient** = stakeholder.
4. Test: add a new item → receive Teams notification.

(You already have detailed step-by-steps in your source; these align with the same pattern.)

Module 8 — OneDrive/SharePoint admin settings, IRM, and (optional) Information Barriers

Why: These govern lifecycle, storage, client sync, and collaboration boundaries.

1. **OneDrive**:
   - Retention (deleted sites) and default storage allocation.
   - **Sync** restrictions: allow only domain-joined devices; block risky file types. [dominiquehermans.com]
2. **Information Rights Management (IRM)** (optional):
   - Enable IRM in classic settings; apply IRM to a library (e.g., "Restricted Files") with download restrictions and expiration. (Extra time may be required before the IRM link appears.)
3. **Information Barriers (IB)** (optional):
   - Explain IB modes (**Open**, **Explicit**, **Owner-Moderated**, **Implicit/Mixed** on OneDrive) and that IB can **block sharing/membership** across disallowed segments in SharePoint/OneDrive. This is a **compliance** feature distinct from audience targeting. [rishandigital.com], [dominiquehermans.com]

Module 9 — Put it all together (checklist & success criteria)

A) Build a reusable taxonomy and light up search & rollups

1. Create term sets (Department, RecordType, Region). [sharepointdiary.com]
2. Create managed metadata site columns; add to content types (Policy, Contract). [mrsharepoint.com]
3. Add content types to libraries across sites.
4. Map crawled → managed properties; set **Refinable/Sortable**. [c-sharpcorner.com]
5. Use **Highlighted Content** on hub/home pages to roll up by managed properties or KQL; enable **audience targeting** as needed. [learn.microsoft.com], [cdm.iamcloud.info]

B) Navigation that scales

- Register hub; associate sites; configure **hub nav** with audience-targeted links; (optionally) configure **global navigation** in app bar. [Refactor_S...Admin_2026 | Word], [pragmaticworks.com], [helpcenter...zensai.com]

C) Large libraries

- Enable **Metadata Navigation and Filtering**; add navigation hierarchies and key filters; confirm auto indexing. [en.ittrip.xyz]

D) People data that works in search (optional)

- Map user profile property to term set; mark privacy = Everyone; map crawled → managed property for people search. [sympraxisc...ulting.com], [mrsharepoint.com]

E) Power Platform

- Build a Power Apps form on a SharePoint list; add a Teams notification flow in Power Automate.

F) Governance

- OneDrive retention/storage/sync; (optional) IRM; **do not delete root**—use **Site Swap** when modernizing the root home site. [m365admin....sontek.net]

Appendix — Minimal "correct order" for the core practice (cheat sheet)

1. **Plan** term sets (names, synonyms). [sharepointdiary.com]
2. Create **Term Store** group/term sets.
3. Create **site columns** (Managed Metadata) bound to the term sets. [mrsharepoint.com]
4. Create **content types**; add the site columns. [sharepointdiary.com]
5. Add content types to **lists/libraries** and tag content.

6. **Search schema**: map crawled → managed; mark refinable/sortable; **reindex** library. [c-sharpcorner.com]
7. Build **HCWP** rollups and (optionally) **audience target** web parts and nav links. [learn.microsoft.com], [cdm.iamcloud.info]

Trainer notes & gotchas

- **Index latency:** Managed property mappings can take minutes to appear; reindex at library/site if needed. [c-sharpcorner.com]
- **Targeting ≠ Security:** Ensure permissions match expectations; audience targeting surfaces content but does not secure it. [cdm.iamcloud.info]
- **Classic to modern:** Avoid new subsites; replace classic roots using **Site Swap** instead of delete. [m365admin....sontek.net]
- **Hub site permissions:** Use -Principals on Register-SPOHubSite to control who can associate sites. [Refactor_S...Admin_2026 | Word]
- **Global nav limits:** Only the **global nav** tab in the app bar is customizable; classic sites don't get the app bar. [helpcenter...zensai.com]
- **IB vs IRM vs Sensitivity Labels:** IB = **who can collaborate**; IRM = **usage restrictions** on content; Sensitivity Labels (outside this lab) = classification + protection. IB impacts SharePoint/OneDrive sharing/membership behaviors. [rishandigital.com], [dominiquehermans.com]

Deliverables you can hand to students

- **Starter topology**: four modern sites + one classic site/subsite.
- **Term Store export** (CSV) for Department/RecordType/Region.
- **Two content types**: AW Policy, AW Contract.
- **A working HCWP rollup** filtered by managed properties.
- **One Power Apps form** and **one Flow** that posts to Teams.
- **Admin settings** screenshots (OneDrive retention/storage/sync; optional IRM).

Reference

1. Planning your SharePoint hub sites - SharePoint in Microsoft 365 | Microsoft Learn